VOLUME 6 SURVEILLANCE

CHAPTER 9 PART 145 INSPECTIONS

Section 2 Inspect a Part 145 Repair Station Manual

Source Basis:

- **Section 145.51, Application for Certificate.**
- **Section 145.109, Equipment, Materials, and Data Requirements.**
- **Section 145.161, Records of Management, Supervisory, and Inspection Personnel.**
- **Section 145.163, Training Requirements.**
- **Section 145.165, Hazardous Materials Training.**
- **Section 145.207, Repair Station Manual.**
- **Section 145.209, Repair Station Manual Contents.**
- **Section 145.211, Quality Control System.**
- **Section 145.215, Capability List.**
- **Section 145.217, Contract Maintenance.**
- **Section 145.221, Service Difficulty Reports.**
- **Part 145, Repair Stations.**

6-1651 REPORTING SYSTEM(S). Use Safety Assurance System (SAS) automation and the associated Data Collection Tools (DCT).

6-1652 OBJECTIVE. This section provides guidance for reviewing the procedures contained in the Repair Station Manual (RSM) required by Title 14 of the Code of Federal Regulations (14 CFR) part 145, § 145.207 and to ensure the contents meet the requirements of § 145.209.

6-1653 GENERAL.

A. Acceptable. The RSM must be acceptable to the Federal Aviation Administration (FAA). This ensures the repair station has procedures for operating in accordance with § 145.207.

B. RSM. The repair station must:

1) Prepare and follow an RSM acceptable to the FAA.

2) Maintain a current RSM in accordance with § 145.207.

3) Make the manual accessible for use by repair station personnel required by part 145 subpart D.

4) Provide to the responsible Flight Standards office the current RSM in a format acceptable to the FAA.

5) Notify the responsible Flight Standards office of each revision of its RSM in accordance with the procedures required by § 145.209(j).

C. Manual Content. The manuals submitted by a certificate holder (CH) or applicant may be separate or may be combined into a single manual. The format should be consistent and all regulatory requirements must be included.

1) The aviation safety inspector (ASI) must ensure that the procedures used in the performance of maintenance, preventive maintenance, or alterations are reflected accurately in the manuals. It is expected that, to fully describe the repair station's inspection/quality system, there will be some procedures that may not be regulatory.

2) There are some recommendations included in this section referenced from Advisory Circular (AC) 145-9, Guide for Developing and Evaluating Repair Station and Quality Control Manuals, which are not required by the regulations. They have been included to assist the inspector and CH/applicant in developing a more complete description of the repair station's overall functions, responsibilities, and quality control (QC) procedures.

D. Original Certification versus Revision. When evaluating a manual as part of an original certification, each entire manual will be submitted prior to certification, the manual will be evaluated to ensure it meets the requirements of part 145. If this task is performed as a revision, only the portion of the manual that is revised must be submitted.

E. Verification. Verify RSM procedures by using a sample of the documents used during maintenance. Examples of documents are: travelers, work orders, inspection sheets, discrepancy sheets, and inspection of articles maintained. Confirm the repair station is following the procedures contained in the RSM.

F. Hazardous Material (HAZMAT). Each repair station that meets the definition of a HAZMAT employer under Title 49 of the Code of Federal Regulations (49 CFR) part 171, § 171.8 must have a HAZMAT training program that meets the training requirements of 49 CFR part 172 subpart H. The HAZMAT training manual should be a separate manual for acceptance under 49 CFR part 172 subpart H and not included as a requirement of the FAA training manual required by 14 CFR § 145.163.

NOTE: Principal inspectors (PI) should ensure that repair stations are aware of the 49 CFR regulations governing HAZMAT transportation. Any HAZMAT training manual received from a repair station must be submitted to the Office of Security and Hazardous Materials Safety (ASH) through the Hazardous Materials Branch Manager (HMBM). The definitions of "HAZMAT employer" and "HAZMAT employee" can be found in 49 CFR § 171.8.

6-1654 REFERENCES, FORMS, AND JOB AIDS.

A. References (current editions):

- Canadian Aviation Regulations (CAR) Parts IV and VII.
- U.S.–Canadian Bilateral Aviation Safety Agreements (BASA)/Maintenance Implementation Procedures (MIP).
- Volume 1, Chapter 3, Section 1, Responsibilities of Aviation Safety Inspectors.
- Volume 2, Chapter 11, Certification of a Part 145 Repair Station:
  - Section 1, Phase 1—Preapplication.
  - Section 2, Phase 2—Formal Application (SAS Business Process Modules 1 and 2).
  - Section 3, Phase 3—Design Assessment (SAS Business Process Modules 2, 3, and 4).
  - Section 4, Phase 4—Performance Assessment (SAS Business Process Modules 2, 3, and 4).
  - Section 5, Phase 5—Administrative Functions.
  - Section 7, Safety Assurance System: Certification of a Part 145 Satellite Repair Station, Amendment to, and/or Transfer of Certificate of the Part 145 Certification.
  - Section 10, Safety Assurance System: Certificating/Renewing/Amending a Part 145 Repair Station Located Outside the Territories of the United States and Not Under a BASA/MIP.
- Volume 14, Chapter 1, Section 2, Flight Standards Service Compliance Action Decision Procedure.
- AC 43-10, United States—Canadian Bilateral Aviation Safety Agreement Maintenance Implementation Procedures.
- AC 145-9, Guide for Developing and Evaluating Repair Station and Quality Control Manuals.
- AC 145-10, Repair Station Training Program.

B. Forms. None.

C. Job Aids. None.

6-1655 RSM REVIEW PROCEDURES. The ASI will ensure the following:

A. Acceptable Formats. Receive the CH's or applicant's manual or revision as required by §§ 145.51, 145.207, 145.209, and 145.211(c). Paper copies and electronic copies are acceptable.

B. Manual or Revision Content. Review the submitted manual or revision to ensure that it meets the regulatory requirements of §§ 145.209 and 145.211. The manual or revision must include the following:

1) An organizational chart that identifies:

a) Each management position with authority to act on behalf of the repair station.

1. The organizational chart required by § 145.209 may identify management positions by title only.

2. This does not eliminate the requirement in § 145.51 for an applicant to submit the names and titles of its management and supervisory personnel at the time of application.

b) The area of responsibility assigned to each management position, which is the area(s) in the repair station that the manager is directly accountable for and maintains decision authority over.

c) The duties, responsibilities, and authority of each management position.

2) Procedures for maintaining and revising the rosters required by § 145.161.

NOTE: Within 5 business-days of the revision, the rosters required by § 145.161 must reflect changes caused by termination, reassignment, change in duties, scope of assignment, or addition of personnel.

3) A description of a repair station's operations describing how the repair station performs maintenance, where it would start, and how it progresses through the entire repair cycle for approval for return to service. Also include:

a) A description of the housing, which may include dimensions, construction method, heating and ventilation systems, lighting, door openings, and physical address.

b) A description of the facilities, including the layout of the shop, hangar, or other work areas.

c) A description of the equipment, tooling, and materials used to perform maintenance.

NOTE: The "description of materials used to perform maintenance" should not be a physical description of the materials, but rather an explanation of the repair station's handling and storage of the materials. If materials require specific environmental controls or cannot be stored next to certain chemicals or solvents, these should be identified. For example, it would not be acceptable to store oxygen equipment near petroleum products.

1. If the repair station does not own the equipment, the manual must include procedures that describe how the repair station will obtain the equipment (e.g., lease or rentals). The manual must also include where repair station personnel will use the equipment, how repair stations will train personnel on the proper use of the equipment, and how the repair station will ensure that it addresses calibration issues, if any, after transporting the equipment.

2. If the repair station chooses to use equipment, tools, or materials other than those recommended by the manufacturer, the manual must include a procedure used by the repair station to determine the equivalency of that equipment, tool, or material.

NOTE: When the repair station is adding a rating or an applicant has applied for certification, all required equipment for the rating it seeks must be in place for inspection by the FAA. This provides the ASI with the opportunity to evaluate its placement and use, and to verify that repair station personnel are trained to operate it.

4) Capability list (CL) procedures used to:

- Revise the CL provided in § 145.215 and notify the responsible Flight Standards office of revisions to the list, including how often the repair station will notify the responsible Flight Standards office of revisions; and
- Develop and perform the self-evaluation required by § 145.215(c) for revising the CL, including the methods and frequency of such evaluations and procedures for reporting the results to the appropriate manager for review and action.

a) A certificated repair station (CRS) with a limited rating may perform maintenance, preventive maintenance, or alterations on an article if listed on a current CL acceptable to the FAA or on the repair station's operations specifications (OpSpecs). If the repair station chooses to use a CL, the RSM must:

- Contain procedures for revising the list and notifying the responsible Flight Standards office,
- Include how often the responsible Flight Standards office will be notified of revision,
- Contain procedures for the self-evaluation required under § 145.215(c) for revising the CL,
- Describe the methods and frequency of such evaluations, and
- Contain procedures for reporting the results to the appropriate manager for review and action.

b) The CL itself may be a separate document or part of the RSM; however, the manual must contain the procedures for revising the list and for performing the self-evaluation.

c) If the repair station elects to maintain a separate CL, it must perform a self-evaluation before adding an article to the CL. The individual(s) performing the self-evaluation should be familiar with the repair station processes and be able to perform an audit to determine compliance with part 145. The self-evaluation procedures in the RSM should ensure that the repair station has:

- The appropriate limited rating;
- Adequate housing and facilities;
- The recommended tools, equipment, and materials;

- Current technical data; and
- Sufficient qualified personnel.

d) The repair station must report the results of the self-evaluation to the appropriate repair station manager for review. If the self-evaluation was satisfactory, the CL may undergo revision. The repair station can submit the revised list and any other necessary technical data with a transmittal document to the PI at the responsible Flight Standards office.

NOTE: Transmittal documents include cover letters, memos, emails, faxes, or any other media acceptable to the responsible Flight Standards office.

e) If the capabilities are maintained on the OpSpecs, each article will be listed by make, model, or manufacturer's name under each limited rating. If the repair station maintains a separate CL, the OpSpecs will indicate that the CH has the authorization to use a CL as revised.

f) A repair station that chooses to use a CL must maintain a current CL acceptable to the FAA, identifying each article by make and model or other nomenclature designated by the manufacturer. A CL should not use the term "all" to denote the make or model. "Series" may describe the model, provided the term does not denote a broad classification that is not well-defined. For example, "Cessna 150 series aircraft" may be an appropriate entry; whereas "Cessna 100 series aircraft" is a broad classification that includes many substantially different models.

g) If the repair station does not maintain or have the necessary tools, equipment, housing, facilities, and trained personnel to perform the required maintenance on the article(s) listed on the CL, delete the article(s) from the CL. However, it does not need to have the tools, equipment, or materials for the scope of work it is authorized to contract out pursuant to its FAA-approved maintenance functions. The exception in the rule "as otherwise prescribed by the FAA" allows a repair station to satisfy § 145.109(a) by way of the FAA-approved maintenance functions under § 145.217. The repair station should ensure the contractor has the necessary equipment, tools, and material for the maintenance functions performed.

NOTE: If the repair station is leasing the tooling or equipment, they should have written evidence that it can obtain the tools and equipment required to maintain the articles on the CL.

5) Procedures for revising the training program and submitting revisions to the responsible Flight Standards office for approval, which should include:

- The title of the person authorized to make a training program revision;
- The method of submitting a revision (e.g., electronic, hardcopy, or disk); and
- A procedure for recording a revision and a method of identifying the revised material or text.

6) Procedures for accomplishing work performed at a location other than the repair station's fixed location, which should contain the following:

a) The title of the person responsible for determining that the location is appropriate for the work performed.

b) The title of the person responsible for initiating such work and assigning the personnel necessary to perform inspections and supervise the work.

c) Procedures for communication between responsible repair station personnel at the fixed location and the maintenance personnel working away from the station. This should include the transfer of parts, supplies, tools/equipment, technical data, and trained personnel.

d) Procedures that maintenance personnel will use when away from the repair station if they deviate from established procedures used at the fixed location. The repair station must ensure that maintenance personnel accomplish all work performed while exercising the privileges of its certificate, per the appropriate maintenance manual and its RSM or Quality Control Manual (QCM). The determination for performing work at another location must meet one of the following requirements:

1. The work is necessary due to a special circumstance, such as a one-time occurrence, as determined by the FAA; or

2. It is necessary to perform such work on a recurring, but not continuous, basis and the RSM includes the procedures for accomplishing maintenance, preventive maintenance, alterations, or specialized services at a place other than the repair station's fixed location.

NOTE: The FAA must make the determination regarding the performance of work at another location prior to the performance of any maintenance, preventive maintenance, or alterations away from the repair station's fixed location unless the manual includes an acceptable procedure.

7) Procedures for performing maintenance, preventive maintenance, and alterations for CHs under 14 CFR parts 121, 125, and 135, and for foreign air carriers or foreign persons operating a U.S.-registered aircraft in common carriage under 14 CFR part 129.

a) The FAA requires that maintenance under a Continuous Airworthiness Maintenance Program (CAMP) be performed in accordance with the operator's manual. It is the operator's responsibility to ensure that the work performed on its behalf is in accordance with its approved maintenance program.

b) The CRS that performs maintenance, preventive maintenance, or alterations for an air carrier or commercial operator that has a CAMP under part 121 or 135 must follow the air carrier or commercial operator's maintenance program or applicable sections of its maintenance manual.

c) A CRS that performs inspections for a CH conducting operations under part 125 must follow the operator's FAA-approved inspection program.

d) A CRS that performs maintenance, preventive maintenance, or alterations for a foreign air carrier or foreign operator operating a U.S.-registered aircraft under part 129 must follow the operator's FAA-approved maintenance program.

e) The FAA may authorize a CRS to perform line maintenance on any aircraft of an air carrier certificated under part 121 or 135, or of a foreign air carrier or foreign operator operating a U.S.-registered aircraft in common carriage under part 129, provided the CRS:

- Has the appropriate ratings to perform the maintenance or preventive maintenance on transport category aircraft;
- Performs such line maintenance in accordance with the operator's manual and approved maintenance program;
- Has the necessary equipment, trained personnel, and technical data to perform such line maintenance; and
- Has OpSpecs that include an authorization to perform line maintenance.

NOTE: A repair station must be appropriately rated to perform line maintenance for an air carrier. This would normally require an airframe rating to accomplish scheduled checks, daily inspections, or the servicing of articles. However, a repair station with the appropriate ratings may accomplish unscheduled maintenance and repairs. This could include avionics facilities limited to avionics functions such as troubleshooting electrical or electronic systems, or replacing defective electronic articles.

8) Procedures for performing maintenance, preventive maintenance, and modifications on Canadian aeronautical products.

a) An FAA CRS may perform maintenance, preventive maintenance, and modifications (with the exception of annual inspections) on a civil aeronautical product under the regulatory control of Transport Canada Civil Aviation (TCCA). The repair station may approve that product for return to service if the repair station complies with the Special Conditions stated in the BASA/MIP between the United States and Canada.

b) In addition to the other requirements specified in the MIP, a repair station performing maintenance, preventive maintenance, or modifications on aircraft operating in commercial air service under TCCA CAR part IV or VII must include in its manual a supplement describing the procedures listed in AC 43-10, Appendix 3, paragraph 3.2, or explain where in the RSM those procedures are described. These procedures must be accepted by the FAA.

9) Procedures for maintaining and revising the contract maintenance information, including the submission of revisions to the responsible Flight Standards office for approval and how often the repair station will notify the FAA of revisions.

a) The FAA must approve the maintenance functions contracted to noncertificated providers.

b) The repair station must maintain a list of each facility that it contracts maintenance functions with, including the type of certificate and ratings (if any) held by each facility.

c) The manual does not need to include the maintenance function list, but the manual should include the location or office where the repair station maintains the list.

NOTE: Maintenance functions are a step or series of steps in the process of performing maintenance, preventive maintenance, or alterations that result in approving an article for return to service. Maintenance functions could be a single step, such as plating a part, or a series of steps, up to and including the overhaul of a component in the process of performing maintenance, preventive maintenance, or alterations that result in approving an article for return to service. It is not the intent of § 145.217(c) to create "virtual repair stations" that provide only an approval for return to service. ASIs must evaluate the amount of work that a repair station desires to contract out versus the work it performs in house.

10) A description of the recordkeeping system used by the repair station to obtain, store, and retrieve the records required by 14 CFR part 43. These records must be in English.

11) Procedures for revising the RSM and notifying the responsible Flight Standards office of revisions to the manual, including how often the repair station will notify the FAA of revisions. The procedure must include:

- The title of the person authorized to make a revision;
- The method of submitting a revision (electronic, hardcopy, disk, etc.);
- A procedure for recording a revision and a method of identifying the revised material or text; and
- A description of the system used to identify and control sections of the RSM.

C. Service Difficulty Reports (SDR) and Suspected Unapproved Parts (SUP). The manual should include the following:

1) Procedures for submitting an SDR. A CRS must report to the FAA within 96 hours after it discovers any serious failure, malfunction, or defect of an article in accordance with § 145.221 and in a format acceptable to the FAA. This is usually in the form of an SDR. If the repair station performs maintenance, preventive maintenance, or alterations for an air carrier, the manual should also contain procedures on how it will notify the operator when submitting reports. The reporting requirement of part 121, § 121.703(d) will be served when an aircraft is scheduled out of service for more than 72 hours due to maintenance, preventive maintenance, or alteration activities. Occurrences and deficiencies must be reported 96 hours after the work on the aircraft is approved for return to service, unless the condition has been reported under another part or section of 14 CFR (e.g., 14 CFR part 21, § 21.3 or § 145.221).

2) Procedures for detecting and reporting SUPs.

D. Compliance Action (CA). Follow the process contained in Volume 14, Chapter 1, Section 2 to identify the root cause that led to any deviations from rules, standards, or procedures, resolve them, and return the repair station to full compliance.

6-1656 TASK OUTCOMES.

A. Conduct Debriefing. Brief the CH on the results. Discuss all deficiencies, CH corrective actions, and FAA actions. The ASI can find instructions for conducting briefings in Volume 1, Chapter 3, Section 1.

B. Compliance and Enforcement Action. If safety issues and/or regulatory noncompliance are identified, follow the process contained in Volume 14, Chapter 1, Section 2 to determine the appropriate FAA compliance or enforcement action.

C. Complete the Task. Follow Volume 10 for completion of the SAS DCT.

6-1657 FUTURE ACTIVITIES. Follow Volume 10 to plan future risk-based surveillance in SAS.

RESERVED. Paragraphs 6-1658 through 6-1668.